Markov Chains Proofs

Wednesday, August 27, 2025 1:37 PM

Perron-Frobenius theorem:

Let A be a square matrix with all non-negative entries and suppose A is irreducible. Then,

(i) spectral radius $|\lambda_{max}(A)|$ is a positive eigenvalue (called the Perron eigenvalue).

(ii) $\exists$ a strictly positive left and right vectors corresponding to the above eigenvalue

(iii) These eigenvectors are unique (up to scaling).

Theorem: If a finite-state Markov chain is irreducible, then it has a unique stationary distribution.

Proof: - From the P-F theorem, $\exists$ a positive eigenvalue that is larger than or equal to the magnitudes of all other eigenvalues.

- Let P be the probability transition matrix.

$$P\mathbb{1} = \mathbb{1} \quad (\mathbb{1}: \text{all ones vector}).$$

So the above eigenvalue ≥ 1

- Consider the Perron eigenvalue λ:

$$\Rightarrow \quad \lambda x = Px \quad \text{for some } x \neq 0$$

$$\Rightarrow \lambda \max_i |x_i| = \max_i |(Px)_i|$$

$$\leq \max_i \sum_j P_{ij}|x_j|$$

$$\leq \max_j |x_j| \max_i \underbrace{\sum_j P_{ij}}_{=1}$$

$$\Rightarrow \quad \lambda \leq 1 \quad \Rightarrow \boxed{\lambda = 1}$$

- By the P-F, $\exists$ v s.t.

$$vP = v \qquad \text{with } v_i \geq 0 \ \forall i$$

Let
$$\mu_i = \frac{v_i}{\sum_{j=1}^{n} v_j}$$

$$\Rightarrow \boxed{\mu P = \mu}$$

$\Rightarrow$ μ is the stationary distribution

□

Theorem: Given two pmfs p, q, define

$$\|p-q\|_{TV} = \frac{1}{2}\|p-q\|_1 = \frac{1}{2}\sum_i |p_i - q_i|$$

If a Markov chain is irreducible and aperiodic, with a stationary distribution μ, then there exists $\rho \in (0,1)$ and $C > 0$ s.t.

$$\|p(k) - \mu\|_{TV} \leq C\rho^k \qquad \forall p(0).$$

Proof: We will rely on the following Lemmas we will prove later:

Lemma (i) If P is irreducible and aperiodic, then $\exists \tilde{M}$ s.t. all entries of P^m are >0 $\forall m \geq \tilde{M}$.

Lemma (ii) The coupling inequality: Let p and q be two pmfs and let X and Y be r.v.'s with a joint pmf p_{XY} s.t. the marginal distributions $p_X = p$ and $p_Y = q$. Then

$$\|p-q\|_{TV} \leq P(X \neq Y)$$

From (i), $P^m_{ij} > 0 \quad \forall i, j$

$$\Rightarrow \quad P^m_{ij} \geq (1-\delta)\mu_j \quad \text{for some } 0<\delta<1$$

$$\Rightarrow \quad P^m_{ij} = (1-\delta)\mu_j + \delta q_{ij} \quad \text{for some } q_{ij} \geq 0$$

Note that
$$\underbrace{\sum_j P^m_{ij}}_{=1} = (1-\delta)\underbrace{\sum_j \mu_j}_{=1} + \delta \sum_j q_{ij}$$

$$\Rightarrow \quad \sum_j q_{ij} = 1 \quad \Rightarrow \quad Q \text{ is a stochastic matrix}$$

$$\Rightarrow \quad P^m = (1-\delta)M + \delta Q \quad \text{where} \quad M = \begin{pmatrix} \mu \\ \mu \\ \vdots \\ \mu \end{pmatrix}$$

$\Rightarrow$ The MC is statistically identical to the following Markov chain X:

- Start with some $p(0)$ and generate $X_m, X_{2m}, X_{3m}, \ldots$ as follows:
  at time $(k+1)m$ w.p. δ pick the next state according to Q with previous state being X_{km}. And w.p. $(1-\delta)$, choose $X_{(k+1)m}$ according to the distribution μ
- At times in between (km) & $(k+1)m$, $k \geq 0$, choose $X_{km+1}, \ldots, X_{(k+1)m-1}$ according $P(X_{km+1}, \ldots, X_{(k+1)(m-1)} \mid X_{km}, X_{(k+1)m})$.
  (Note that we don't have to do this, it is just a construction for the purpose of the proof.)

- Motivated by the above construction, let us consider two Markov chains Y & Z, with initial distributions ν and μ (the stationary distribution). Both Y and Z are generated as we did X above; however whenever the Ber(δ) r.v. asks to choose a state according to μ, say at some time km for some $k \geq 1$, then choose a state according to μ and set both X_{km}, Y_{km} to be the same state. The two MCs are said to couple at this time. From then on, evolve just one of them, say Z, and let $Y_t = Z_t$ $\forall t > km$

- Therefore, we have created two MCs Y & Z that evolve according to P and once they couple, then stay identical from then on.

- Let $p_Y(k)$ and $p_Z(k)$ be the prob. distributions associated with X and Y, respectively, at time k. Then, by the coupling inequality Lemma (ii),

$$\|p_Y(t) - p_Z(t)\|_{TV} = \|\nu P^t - \mu\|_{TV}$$

$$\underset{\substack{\uparrow \\ \text{Lemma (ii)}}}{\leq} P(Y_t \neq Z_t) \leq \delta^{\lfloor \frac{t}{m} \rfloor} = \delta^{t/m}\left(\delta^{\lfloor \frac{t}{m} \rfloor - \frac{t}{m}}\right)$$

$$\leq \underbrace{\frac{1}{\delta}}_{C} \underbrace{(\delta^{1/m})}_{\rho}{}^{t}$$

□

Proof of Lemma (i)

Since the Markov chain is irreducible, there exists an $m_{ij} \geq 1$ s.t.

$$p_{ij}^{(m_{ij})} > 0 \qquad \forall i, j.$$

Further since the Markov chain is aperiodic, $\exists$ N_i s.t. $\forall n \geq N_i$,

$$p_{ii}^{(n)} > 0 \qquad \forall i$$

Since the number of states is finite, this implies

$$p_{ii}^{(n)} > 0 \qquad \forall n \geq N \text{ (some N)} \quad \forall i$$

For all ~~each~~ n:

$$p_{ij}^{(n+m_{ij})} = \sum_s p_{is}^{(n)} p_{sj}^{(m_{ij})}$$

$$\geq p_{ii}^{n} p_{ij}^{m_{ij}} > 0$$

Let $M = \max_{i,j} m_{ij}$. Then, for any $n+M$, $n \geq N$, $p_{ij}^{(n+M)} > 0$

$$\Rightarrow \quad p_{ij}^{(m)} > 0 \qquad \forall m \geq N+M.$$

□

Proof of Lemma (ii)

$$P(X \neq Y) = 1 - P(X=y)$$

$$= 1 - \sum_i \underbrace{P(X=i, Y=i)}_{\leq P(X=i) \text{ and } \leq P(Y=i)}$$

$$\geq 1 - \sum_i \min(p_i, q_i)$$

$$\|p-q\|_{TV} = \frac{1}{2}\sum_i |p_i - q_i|$$

$$= \frac{1}{2}\sum_i \left(p_i + q_i - 2\min(p_i, q_i)\right)$$

$$= \frac{1}{2}\left(1+1-2\sum_i \min(p_i, q_i)\right)$$

$$= 1 - \sum_i \min(p_i, q_i)$$

$$\Rightarrow \quad \|p-q\|_{TV} \leq P(X \neq Y)$$

$\forall$ (X, Y) with marginals p and q

□

WLLN

Let X be an irreducible, aperiodic finite state-space MC.

Theorem:
$$\frac{1}{N}\sum_{t=0}^{N-1} f(X_t) \xrightarrow{P} \sum_i \mu_i f(i) \quad \text{for all functions } f.$$

where μ is the stationary distribution

Proof:

Let Y be another MC coupled with X as in the proof of the previous theorem.

$$E\left(\left|\frac{1}{N}\sum_{t=0}^{N-1} f(X_t) - \underbrace{\frac{1}{N}\sum_{t=1}^{N} f(Y_t)}_{\substack{\text{This, in expectation,} \\ = \sum_i f(i)\mu_i}}\right|\right)$$

$$= \frac{1}{N} E\left(\sum_{t=0}^{N-1} |f(X_t) - f(Y_t)|\right)$$

$$\leq \frac{2 f_{max}}{N} E(\min(T, N)), \quad \text{where T is the coupling time}$$

$$\leq \frac{2 f_{max}}{N} E(T)$$

$$= \frac{2 f_{max}}{N}\left[(1-\delta) + 2\delta(1-\delta) + 3\delta^2(1-\delta) + \cdots\right]$$

$$= \frac{f_{max}}{N}\frac{1}{(1-\delta)}$$

$$\Rightarrow P\left(\left|\frac{1}{N}\sum_{t=0}^{N-1} f(X_t) - \underset{X_0 \sim \mu}{E}[f(X_0)]\right| \geq \varepsilon\right)$$

$$\leq \frac{1}{\varepsilon} E\left(\left|\frac{1}{N}\sum_{t=0}^{N-1} f(X_t) - E_{X_0 \sim \mu}(f(X_0))\right|\right)$$

$$\leq \frac{1}{\varepsilon}\frac{f_{max}}{N(1-\delta)} \to 0 \quad \text{as } N \to \infty.$$

□